

Week 5 Interim — Feasibility Memo Outline

Dataset: Yale EMLC ED Triage & Admission Prediction

Mercer General Hospital — Clinical AI & Innovation Unit

1. One-Sentence Verdict

This dataset is feasible to build on, with caveats around missingness in clinical variables, single-site US origin, and class imbalance in the target variable.

2. Dataset Summary

Property	Detail
Source	Yale New Haven Health System, Connecticut, USA
Time period	March 2014 – July 2017 (approximately 40 months)
Size	~560,000 adult ED visits, 225 features per visit
Target variable	ESI triage level (1–5) and/or admission vs discharge (binary)
Exclusions	Paediatric patients, non-clear dispositions (transfer, AMA, eloped)
De-identification	Confirmed. PHI removed. Published under peer review (Hong et al., 2018, PLOS ONE)
Feature groups	Demographics, triage vitals, chief complaints (top 200, binary-encoded), hospital usage history, past medical history

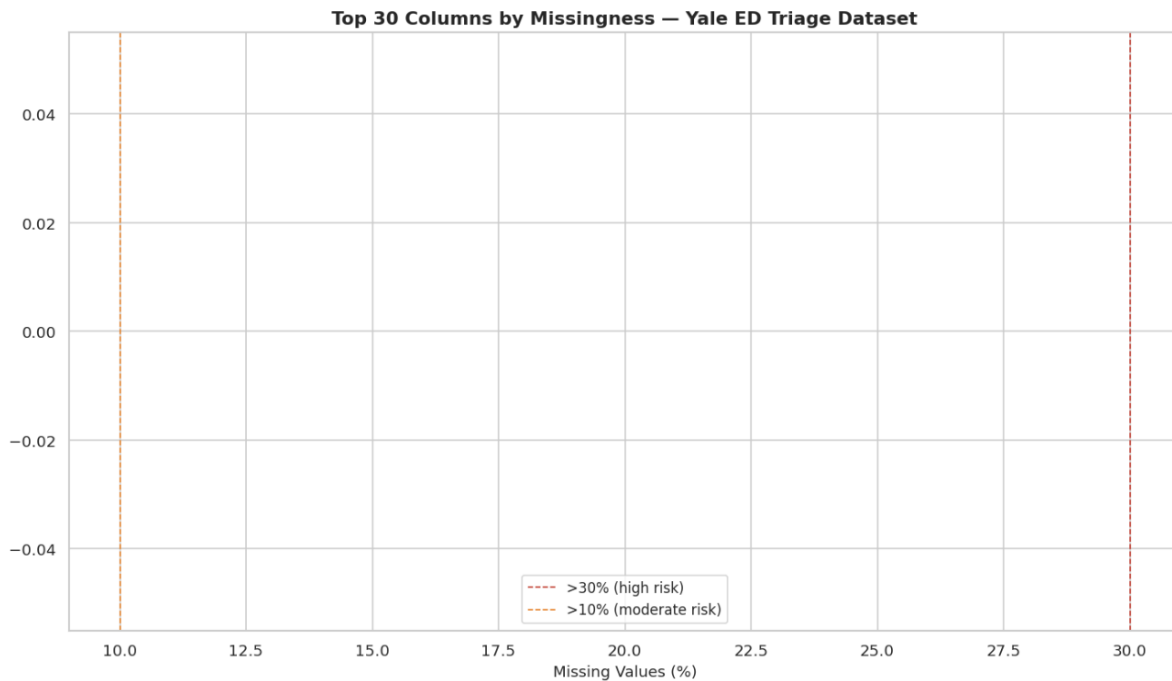
Category	Examples	Count (approx.)
Demographics	Age, gender, ethnicity, language, insurance, employment, marital status	~10
Triage vitals	Temperature, blood pressure, pulse, respiratory rate, O2 saturation, pain score, ESI	~8
Chief complaints	Binary-encoded top 200 chief complaints (chest pain, dyspnoea, etc.)	~200
Hospital usage	ED visits in past year, admissions in past year, prior visit disposition	~5
Past medical history	Prior diagnoses, procedures, medications listed at time of encounter	~varied

3. Top 3 Quality Concerns

Concern 1 — Missingness in clinical variables

Several vital sign columns have missingness rates above 30%, including variables clinically important for triage modelling such as oxygen saturation and pain score. Imputing these without

a defensible clinical rationale risks introducing noise. Dropping them reduces the predictive feature set.

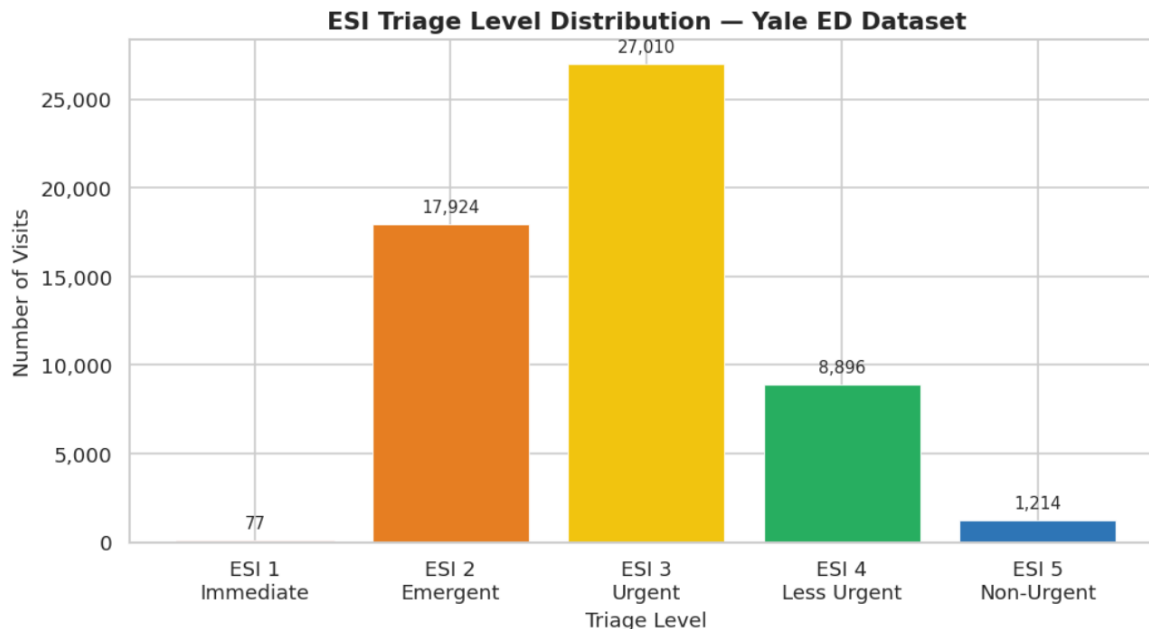


Concern 2 — Single-site US origin: distribution shift risk

All visits originate from one academic centre in Connecticut. Patient demographics, disease burden, and triage workflows differ substantially from a Caribbean public hospital. A model trained here may not generalise to Mercer General without prospective site-specific validation (De Freitas et al., 2020).

Concern 3 — Class imbalance in the target variable

ESI levels 1 (immediate) and 5 (non-urgent) are heavily underrepresented. A model trained without addressing this will underperform for the most critical patients — exactly the group where errors carry the greatest clinical consequence.



4. Top 3 Reasons to Proceed

Reason 1 — Scale

560,000+ visits provides enough volume to train robust models and perform meaningful subgroup analyses even after aggressive filtering for data quality.

Reason 2 — Feature richness

225 features covering all standard triage vitals and 200 binary-encoded chief complaint categories represents the full clinical picture available to a triage nurse at the point of patient encounter.

Reason 3 — Peer-reviewed provenance

The dataset accompanies a published PLOS ONE study (Hong et al., 2018, DOI: 10.1371/journal.pone.0201016). Cleaning, de-identification, and feature engineering are fully documented and satisfy clinical governance standards for hospital research use.

5. Caveats

- Any model must be prospectively validated at Mercer General before clinical use. Yale performance figures cannot be assumed to transfer.
- Demographic features (insurance, employment, language) may encode socioeconomic bias and must be handled carefully to avoid proxy-variable failures of the type documented by Obermeyer et al. (2019).
- Dataset covers adult patients only. No paediatric visits. Extension to under-18 populations requires a separate data source.

- Chief complaints reflect the top 200 presentations at Yale. Presentations common in Caribbean settings but rare in Connecticut may be absent or underrepresented.

6. Interim Visualisations

Produced in the accompanying Jupyter notebook. Full-resolution files committed to \docs on GitHub.

Plot	Description	File
Missingness bar chart	Top 30 columns by % missing, colour-coded by risk level	week5_missingness_barchart.png
Missingness heatmap	Core variable missingness across 500-row sample	week5_missingness_heatmap.png
ESI distribution	Patient count per ESI level 1–5	week5_esi_distribution.png
Age distribution	Patient age histogram with median line	week5_age_distribution.png
Ethnicity distribution	Top 10 ethnicity/race categories	week5_ethnicity_distribution.png
Chief complaints	Top 20 chief complaints by frequency	week5_chief_complaints.png
Vital signs boxplot	Outlier check across key vital sign columns	week5_vitals_boxplot.png